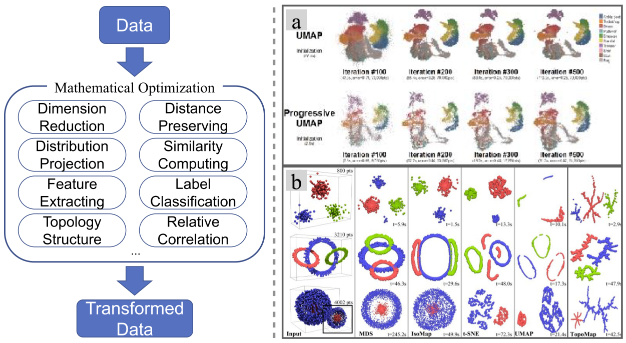

Fig. 3. The basic methodological paradigm of *Data Enhancement & Transformation* and two examples in *Dimension Reduction*. (a) Compared with previous work, The progressive UMAP [33] performs better in time. (b) Compared with other classical methods, TopoMap [32] can preserve more topology structure of data.

space as much as possible. There are many proposed dimension reduction techniques such as PCA, MDS, and t-SNE. However, dimension reduction inevitably results in the loss of information. Most papers employ unconstrained optimization models. But they different objective functions to preserve certain information that users are concerned such as principal component, relative correlation, clusters, topology correlation, or specific data information.

A classical dimension reduction method is Principal Component Analysis (PCA), which first projects data in the direction of the greatest variance. Due to the requirement of data comparison and dynamic streaming data, many variants of PCA are formulated to compare the data item with clusters [21] and compute data positions incrementally by mathematical optimization [22].

Conserving relative distance aims to analyze relative correlations among data items. Multi-dimension Scaling (MDS) and its extended algorithm are commonly used for dimension reduction by optimizing the objective function of distance. Apart from preserving the relative distance, the variants of MDS balance multiple viewpoints by integrating them into an objective function [23], consider order discord by depth penalty term [24], and cluster points by density-based method [25]. Other works utilize results of MDS to cluster elements [25], [26] due to the characteristics of distance-preserving.

t-SNE has gained much attention because it can preserve distribution in low-dimensional space, and has many variants with modification of nonlinear objective function (KL-divergence). The variants integrate penalty terms of time variance to project temporal data [27], combine clusters to balance readability [28], and minimize vertical distance terms for bipartite graph exploration [29].

One of the critical problems in dimension reduction is preserving the specific structure or pattern in the process of projection. Mathematical optimization could help enhance patterns by the dissimilarity metric [30], project local structure with user knowledge [31], and preserve topology structure [32]. For example, retaining topological features [32] is considered when projecting high-dimensional data, as shown in Fig. 3(b). Due to their time cost and complexity, many works aim to improve the time efficiency for UMAP by a progressive method [33] and interactivity of local affine multidimensional projection [34], as shown in Fig. 3(a). Nonetheless, dimension reduction techniques for preserving topology relationships garner attention in geographic maps [30], [35]. These works utilize mathematical optimization models to encode the data dissimilarity [30] and enhance the metro line [35].

Many works project high dimension to low dimension in radial direction such as star coordinate graph [20], [36], [37], [38]. Rubio et al. [36], [38] aims to enhance data estimation in radial axes graph and an orthographic star coordinate is purposed to decrease distortions by Lehmann and Theisel [37]. SolarView [39] formulates a low-distortion radial embedding problem as TSP to visualize entities and relations in the radial graph.

B. Data Refinement

Data Refinement is a process of filtering, selecting, and clearing data that is critical and salient. it tends to abstract or transform raw data into a reasonable form to gain insights while retaining original information as much as possible.

Sampling is a commonly used technique to refine and abstract partial data, such as streaming data [40], scattered data [41], and multi-dimensional data [42]. Sampling data can improve accuracy but can be time-consuming. Xie et al. [42] utilize LP to accelerate sampling when selecting the initial position of the MCMC method.

Sorting is a rearrangement and transformation method of data refinement, which can be viewed as a mathematical optimization process [43], [44] to find the best order of data arrangement. This problem could be considered as a linear assignment problem and formulated as an LP model [43], [45] to determine the best order. GraphScape [46] and AutoClips [47] utilize LP models to arrange the chart order to generate a graph sequence or video.

Extracting is to detect and preserve significant information and neglect redundant data records, which could refine and compress data. Minimum Description Length (MDL) is a prevailing method of data compression in information theory. MDL balances the description length of data and complexity of the model for data summary, which employs MDL to aggregate nodes [48] and summarize groups of temporal event data [49]. Other works like Guo et al. [18] detect latent clusters and extract key information in temporal event data by Canonical Polyadic Tensor Decomposition algorithm. By discarding redundant data and preserving valid data, data refinement can further improve the performance of detecting significant information.

Certain researchers refine and transform data to recommend significant information such as [50], [51], [52]. These works project selected points to high-dimensional space to recommend chart [50], recommend neurons by a subset selection problem [52], [53], and refine event data based on word embedding [54].

In scientific visualization, mathematical optimization could help compute and extract features before visual encoding and rendering. Many works aim at tracking features by weight graph matching [55], computing geometric shapes of potential vorticity [56], and extracting vortices in vector fields [57]. One of the critical problems in vector field computation is the coordinate transformation. Mathematical optimization models could help